

Md. Mahdi Hasan and Badhon Rozario

A Thesis in the Partial Fulfillment of the Requirements
for the Award of Bachelor of Computer Science and Engineering (BCSE)



Department of Computer Science and Engineering
IUBAT School of Computer Science and Engineering
IUBAT—International University of Business Agriculture and Technology

Spring 2026

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Md. Mahdi Hasan and Badhon Rozario

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Letter of Transmittal

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Abstract

Your thesis abstract must be limited to 500 words and should serve as a concise yet comprehensive summary of your entire work, capturing the motivation, problem, approach, results, and conclusion in a clear flow. Although brief, the abstract must accomplish almost as much as the full thesis, so ensure all essential components are included: first, highlight the motivation by explaining the importance of your work, the difficulty of the area, and the potential impact if successful; second, present the problem statement by clearly identifying the issue you are addressing and defining the scope of your work, whether generalized or specific, noting that in some cases the problem may precede the motivation if its importance is already understood; third, describe your approach by summarizing how your proposal solves or makes progress on the problem, mentioning whether you used simulations, analytic models, prototype construction, or field data analysis; fourth, present your results effectively to demonstrate how your proposal performs better; whether faster, smaller, cheaper, or more efficient than existing solutions; and finally, conclude by stating the implications of your research and reinforcing its significance in advancing knowledge or solving real-world problems. Write in formal, precise language, avoid unnecessary jargon, and ensure the abstract stands alone as a self-contained representation of your thesis.

Acknowledgments

We would like to express our deepest gratitude to everyone who supported us throughout the completion of this thesis titled "Personalized Medical Recommendation System Using Explainable AI (XAI)". This journey has been both challenging and rewarding, and it would not have been possible without the guidance and encouragement of several individuals and institutions.

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Table of Contents

Letter of Transmittal	iii
Student's Declaration	iv
Supervisor's Certification	v
Abstract.....	vi
Acknowledgments	vii
List of Figures.....	xi
List of Tables	xii
Chapter 1. Introduction.....	1
1.1 Background	1
1.2 Overview.....	1
1.3 Motivation.....	2
1.4 Problem Statement.....	2
1.5 Objectives	2
1.6 Research Questions	2
1.7 Scope and Limitations.....	3
1.8 Significance of the Research.....	3
1.9 Thesis Outline	3
1.10 Headings	3
Chapter 2. Literature Review	5
2.1 Section Title	8
2.2 B-Head Style	8

2.3 C-Head Style and Accompanying Paragraph	9
2.3.1 Body Text with C-Head	9
2.4 Numbered List Style	9
2.5 Bulleted List Style.....	10
2.6 Referencing	10
2.6.1 Referencing in Text.....	11
2.6.2 The Reference List	12
Chapter 3. Research Methodology	22
3.1 Figures.....	24
3.1.1 Figure Style	25
3.1.2 Figure Description Style	25
3.2 Tables	26
3.2.1 Table Title Style.....	26
3.2.2 Table Cell Style.....	26
3.2.3 Table Description Style.....	27
Chapter 4. Results and Discussion	28
4.1 Presenting Results	28
4.2 Interpreting Results in Discussion	29
4.3 Connecting to Research Objectives	29
4.4 Addressing Limitations	29
4.5 Highlighting Contributions and Implications	29
4.6 Structuring the Chapter	30
4.7 Thesis Title Style	30

4.8 Department School University Style	30
Chapter 5. Conclusion	31
5.1 Restating the Research Problem and Objectives	31
5.2 Summarizing Key Findings	31
5.3 Highlighting Contributions	32
5.4 Discussing Limitations.....	32
5.5 Suggesting Future Work	32
References	33

List of Figures

Figure 3.1 Example Figure Title..... 25

Figure 3.2 Example Figure Title2..... 26

List of Tables

Table 2.1 An example to understand how to prepare the literature review chapter	6
Table 2.2 In Text Citation Style.....	11
Table 2.3 IEEE Referencing Style quick guide	14
Table 3.1 Table Title.....	27

Chapter 1. Introduction

The integration of Artificial Intelligence (AI) into the healthcare domain has catalyzed a transformative shift in how medical services are delivered and managed. In recent years, the exponential growth of digital health records and synthetic datasets has provided an unprecedented opportunity to develop intelligent systems capable of assisting clinicians in decision-making processes [1]. Within this landscape, personalized medical recommendation systems have emerged as a critical tool, aiming to tailor healthcare interventions to the specific needs of individual patients based on their demographic profiles, clinical histories, and treatment patterns. Our research focuses on this intersection of deep learning and personalized medicine, seeking to enhance the precision of procedural recommendations through advanced computational architectures.

Despite the high predictive accuracy achieved by contemporary deep learning models, their adoption in clinical settings is often hindered by their "black-box" nature. In the medical field, where decisions directly impact human lives, the rationale behind a recommendation is as vital as the recommendation itself [2]. Consequently, Explainable AI (XAI) has become a paramount requirement for fostering trust between AI systems and healthcare practitioners. In this thesis, we address this challenge by developing a recommendation framework that not only achieves robust performance but also provides interpretable insights into its decision-making logic using global and local explanation techniques [3].

Our research methodology centers on the development of a Hybrid Multi-Layer Perceptron and Graph Attention Network (Hybrid MLP-GAT) model. By constructing a patient similarity graph, we leverage both individual patient attributes and relational

connectivity to predict medical procedures. This approach allows us to capture complex dependencies within healthcare data that traditional models might overlook. The primary objective of our work is to demonstrate that combining structured feature learning with graph representation learning can significantly improve recommendation accuracy while maintaining a high degree of transparency through XAI frameworks like SHAP and LIME.

In the following sections, we provide a comprehensive overview of the research landscape. We begin by discussing the background of personalized medicine and the current state of GNNs in healthcare. Subsequently, we outline the motivation behind our study, define the specific research problem we aim to solve, and present our objectives and methodology. This chapter concludes by highlighting the expected contributions of our work to the field of explainable healthcare analytics, setting the stage for the detailed technical implementation presented in the subsequent chapters.

1.1 Background

The landscape of modern healthcare is currently undergoing a radical transformation, driven by the rapid integration of Artificial Intelligence (AI) and computational intelligence. As the volume of healthcare data generated through electronic health records (EHRs), diagnostic imaging, and synthetic generation tools like Synthea continues to grow, the need for advanced analytical frameworks has become more pronounced [1]. In our research, we observe that AI is no longer merely a supplementary tool but a core component in the modernization of medical infrastructure. AI's ability to parse through millions of clinical records allows for the identification of patterns that are often imperceptible to the human eye, facilitating early diagnosis, predictive risk assessment, and efficient clinical workflows.

Healthcare decision-support systems (HDSS) represent a significant evolution in this digital journey. Traditionally, these systems relied on rule-based engines where medical knowledge was manually encoded by practitioners. However, such systems often lacked the flexibility required to handle the inherent complexity and variability of patient-specific clinical pathways. With the advent of machine learning, decision-support systems have transitioned toward data-driven paradigms. These modern systems analyze historical patient data to assist clinicians in selecting the most effective interventions [2]. During our implementation, we recognized that the effectiveness of these systems hinges on their ability to integrate multifaceted data points—including demographics, previous encounters, and medication histories—into a cohesive predictive framework.

Recommendation systems, which achieved immense success in the commercial sectors of e-commerce and digital media, are now being adapted for high-stakes medical applications. Unlike consumer-grade recommendations, medical recommendation systems must operate under stringent accuracy and safety requirements. The objective shifts from suggesting a product to suggesting a clinical procedure or medication that aligns with a patient's unique physiological profile. In our thesis, we focus specifically on personalized medical procedure recommendations. We believe that personalization is the key to reducing medical errors and optimizing treatment outcomes. By moving away from a 'one-size-fits-all' approach, we can leverage graph-based relational data to understand how similarity between patients can inform better procedural choices.

Machine learning applications in medicine have expanded across various specialties, from oncology to cardiology. Supervised learning models, such as Multi-Layer Perceptrons (MLP), have shown remarkable proficiency in classification tasks where feature-to-label

mappings are well-defined. However, healthcare data is rarely independent and identically distributed. Patients are often connected through similar symptoms, demographic backgrounds, or institutional settings. This relational aspect led us to explore Graph Neural Networks (GNNs), specifically Graph Attention Networks (GATs). GATs allow our model to assign varying degrees of importance to different neighbors in a patient similarity graph, effectively capturing the contextual nuances of healthcare delivery [3].

One of the most significant barriers to the clinical adoption of advanced machine learning models is the issue of interpretability. While complex models like the Hybrid MLP-GAT we developed offer superior predictive performance—achieving an accuracy of 81.58% in our experiments—they are often criticized as 'black boxes.' In the medical domain, a recommendation without a justification is frequently unusable. This is where Explainable AI (XAI) becomes critical. XAI aims to bridge the gap between high-performance computing and human intuition by providing insights into why a model reached a specific conclusion. In our research, we have integrated both global and local explainability methods to ensure that our system is transparent and trustworthy for medical professionals.

Global explainability, which we implemented using SHAP (SHapley Additive exPlanations), provides an overview of which features are most influential across the entire dataset. This allows us to validate if the model is learning clinically relevant patterns—such as the high importance of 'unique_procedure' and 'AGE' that we observed in our findings. Simultaneously, local explainability, implemented through LIME (Local Interpretable Model-Agnostic Explanations), focuses on individual predictions. By explaining why a specific procedure was recommended for a specific patient, we provide a level of granularity that supports personalized clinical decision-making.

As we narrow the focus of our research, we identify a specific gap in the current literature: the need for a hybrid approach that combines structured feature learning with relational graph learning while maintaining strict explainability. Many existing systems focus either on the patient's individual attributes or broader relational data, but rarely in a unified, interpretable architecture. Our research addresses this by utilizing a Hybrid MLP-GAT model. The MLP component processes individual patient features like income, gender, and encounter counts, while the GAT component captures the similarity-based dependencies between patients.

In conclusion, the background of our thesis is rooted in the necessity for more accurate, personalized, and, most importantly, explainable medical recommendation systems. By leveraging the power of graph-based deep learning and the transparency of XAI, we aim to provide a framework that not only improves procedural recommendation accuracy but also aligns with the ethical and practical requirements of the healthcare industry. Our work serves as a foundation for future clinical decision-support tools that empower practitioners with data-driven, yet fully interpretable, medical insights.

1.2 Overview

The personalized medical recommendation system developed in our research is a multifaceted framework designed to provide high-precision clinical procedure suggestions while maintaining transparency through Explainable AI (XAI). In this section, we present a detailed overview of the system architecture, the operational workflow, and the integration of graph-based deep learning with interpretability methods. The primary goal of our system is to

transform raw healthcare records into actionable, explainable insights that can assist practitioners in identifying the most appropriate medical interventions for patients [4].

1.2.1 System Workflow and Input Processing

Our implementation begins with the ingestion of structured healthcare datasets containing demographic information, medical encounters, and historical procedure records. During the input processing phase, we perform extensive data aggregation to transition from a record-centric to a patient-centric view. We consolidate individual patient attributes—such as age, gender, and income—with aggregated clinical statistics, including the total number of encounters, unique procedure counts, and healthcare expenditure. To handle the high variability of clinical codes, we focused our recommendation engine on the top-5 most frequent procedure categories, ensuring model stability and statistical significance. This processed feature set, comprising 16 engineered features including income-age interactions and log-transformed financial metrics, serves as the input for our neural architecture.

1.2.2 Recommendation Generation and Graph Construction

A core innovation in our research is the construction of a patient similarity network. Unlike traditional recommendation systems that treat patients as isolated entities, we utilize a K-Nearest Neighbors (KNN) algorithm to establish relationships between patients based on their normalized feature profiles. Each patient is represented as a node in a graph, and edges are established with the k most similar patients, creating a rich relational structure where nodes with similar clinical and demographic backgrounds are interconnected.

For recommendation generation, we developed a Hybrid Multi-Layer Perceptron and Graph Attention Network (Hybrid MLP-GAT). The MLP component of our architecture is responsible for deep feature extraction from individual patient attributes, while the GAT component leverages an attention mechanism to weigh the influence of neighboring nodes in the similarity graph. This hybrid approach allows the model to learn both the specific characteristics of the target patient and the shared clinical patterns found within their neighborhood. Through iterative training and hyperparameter optimization—specifically tuning k-neighbors, hidden layer dimensions, and attention heads—we achieved a peak predictive accuracy of 81.58%, outperforming the baseline GAT model which attained 77.37%.

1.2.3 Explainable AI Integration

Recognizing that accuracy alone is insufficient for clinical adoption, we integrated a comprehensive XAI framework into our system. Our approach employs a surrogate model strategy to provide both global and local interpretations. To understand the global behavior of our recommendation system, we utilized SHAP (SHapley Additive exPlanations). Our SHAP analysis identified that 'unique_procedure' history and patient 'AGE' are among the most critical drivers of the recommendation process across the entire population [5].

Furthermore, to support patient-specific decision-making, we implemented LIME (Local Interpretable Model-Agnostic Explanations). LIME provides a granular view of individual predictions, identifying which specific features—such as recent healthcare costs or demographic factors—positively or negatively influenced a particular procedure recommendation. By combining these global and local insights, we ensure that our system

operates as a 'glass-box' framework, where clinicians can verify the logic behind each AI-generated suggestion.

1.2.4 Final Architecture

The final architecture of our system represents a unified pipeline: it starts with synthetic healthcare data preprocessing, moves through a similarity-based graph construction phase, utilizes a hybrid deep learning model for classification, and concludes with an XAI-driven interpretability layer. This integrated design ensures that our system is not only robust in its predictive capabilities but also adheres to the emerging ethical and regulatory standards for transparent AI in medicine. Through this methodology, we provide a scalable framework that can be adapted to various clinical environments to enhance the quality of personalized patient care.

1.3 Motivation

The motivation behind our research is rooted in the critical need for advanced decision-support tools that can navigate the increasing complexity of modern clinical pathways. As healthcare systems transition toward data-driven models, a significant challenge remains: the effective personalization of patient care without compromising the transparency of the underlying technology. In our research, we observed that while standard machine learning models can process large volumes of synthetic healthcare data, they often fail to capture the relational nuances that exist between similar patient profiles. This gap in the existing

literature motivated us to develop a system that leverages both individual characteristics and collective clinical patterns.

One of the primary motivations for this thesis is persistence of trust issues regarding AI adoption in medicine. Despite achieving high accuracy, many neural architectures are perceived as opaque by healthcare practitioners [6]. During our implementation, we recognized that for a medical recommendation system to be viable in a clinical setting, it must provide a clear rationale for its suggestions. This requirement led us to integrate Explainable AI (XAI) as a fundamental pillar of our architecture. By incorporating SHAP for global feature importance and LIME for local patient-level explanations, we aim to bridge the 'interpretability-performance' trade-off. Our motivation stems from the belief that 'glass-box' models are essential for fostering the clinical confidence necessary to assist doctors in selecting appropriate procedures, such as 'Medication Reconciliation' or 'Assessment of health and social care needs,' which were prevalent in our dataset.

Furthermore, the need for deep personalization in medical interventions served as a strong catalyst for our study. Standard recommendation engines often neglect the connectivity between patients who share similar symptoms, demographic backgrounds, or socioeconomic factors. In our system, we addressed this by constructing a patient similarity graph using the K-Nearest Neighbors (KNN) algorithm. Our motivation was to move beyond isolated feature analysis and instead utilize the power of Graph Attention Networks (GAT) to weigh the influence of similar patients. This approach allowed us to capture complex, non-linear relationships within healthcare data, leading to a notable improvement in accuracy over traditional baseline models.

Our research is also motivated by the specific challenge of class imbalance and complexity in procedural healthcare records. In our initial analysis of the 36 available procedure classes, we identified significant variance in frequency. This motivated our decision to focus on the top 5 most frequent procedure categories, ensuring that our hybrid model—achieving 81.58% accuracy after tuning—provides stable and reliable recommendations. By optimizing hyperparameters like hidden dimensions and attention heads, we demonstrated that a carefully tuned hybrid architecture could significantly enhance the precision of personalized medical advice.

Ultimately, the motivation of this thesis is to contribute to a future where AI-driven healthcare is both highly accurate and fully explainable. We believe our Hybrid MLP-GAT framework, combined with dual-layered XAI analysis, offers a blueprint for the next generation of clinical decision-support systems. By prioritizing both relational learning and model transparency, we aim to provide a tool that not only predicts the next medical procedure with high confidence but also empowers healthcare professionals with the underlying 'why' behind every recommendation.

1.4 Problem Statement

Despite the significant advancements in clinical decision-support systems, two primary challenges persist in the development of automated medical recommendation frameworks: the lack of deep personalization and the inherent opacity of high-performance predictive models. Current solutions in the healthcare domain often rely on either standard statistical methods or traditional deep learning architectures that treat patients as independent, isolated data points. We have observed that these methodologies fail to account for the complex,

relational dependencies between patients who share similar clinical histories, demographic traits, and socioeconomic backgrounds. This oversight limits the system's ability to provide truly personalized recommendations, as it ignores the wealth of information embedded in the collective patterns of similar patient cohorts [7].

Furthermore, the "black-box" nature of existing state-of-the-art models, such as standard Multi-Layer Perceptions or uninterpreted Graph Neural Networks, creates a substantial barrier to clinical trust. In a medical context, a recommendation system that lacks transparency is often dismissed by practitioners because it cannot provide a verifiable rationale for its output. As evidenced by our initial analysis, while models like the baseline GAT can achieve respectable accuracy (77.37%), they do not inherently offer insights into why a specific procedure—such as a 'Medication Reconciliation' or 'Hospice care'—is suggested for a particular individual. This lack of interpretability poses ethical and operational risks, particularly when AI-driven suggestions contradict or supplement human judgment.

In our research, we identify the core problem as the absence of a unified framework that simultaneously captures individual patient features and relational graph-based connectivity while providing both global and local interpretability. Specifically, there is a need for a model that can process multifaceted inputs—including 16 engineered features like 'income_log' and 'age_income_interaction'—while leveraging the similarity-based relationships established through a K-Nearest Neighbors (KNN) graph. The problem we address is twofold: first, the performance gap caused by excluding relational patient data in recommendation tasks, and second, the transparency gap that prevents clinicians from understanding the driving factors behind automated medical advice.

Consequently, our thesis addresses this combined problem by developing and optimizing a Tuned Hybrid MLP-GAT model. By integrating this hybrid architecture with a dual-layered XAI framework using SHAP and LIME, we seek to demonstrate that it is possible to achieve superior predictive accuracy (81.58%) without sacrificing the interpretability required for safe clinical implementation. This research aims to provide a solution that balances the technical demands of graph-based representation learning with the practical necessity of explainable, patient-centric healthcare analytics [8].

1.5 Objectives

1.5.1 Overall Objective

The primary objective of this research is to design, implement, and evaluate a robust, personalized medical procedure recommendation system that leverages the relational power of Graph Neural Networks and the feature extraction capabilities of Multi-Layer Perceptions, while ensuring clinical transparency through the integration of global and local Explainable AI (XAI) frameworks.

1.5.2 Specific Objectives

To achieve the overall objective, we have established the following specific goals:

1. To preprocess and aggregate multi-source synthetic healthcare data into a patient-centric feature matrix containing demographic, clinical, and financial attributes.
2. To construct a patient similarity network using the K-Nearest Neighbors (KNN) algorithm to capture the relational dependencies between patients with similar clinical profiles.

3. To develop a Hybrid MLP-GAT architecture that simultaneously learns from individual patient features and the graph-structured connectivity of the similarity network.
4. To perform comprehensive hyperparameter optimization, specifically tuning the number of graph neighbors, attention heads, and hidden dimensions to maximize recommendation accuracy.
5. To evaluate the performance of the proposed hybrid system against a baseline Graph Attention Network (GAT) using standard classification metrics, including accuracy, F1-score, and confusion matrices.
6. To implement global explainability using SHAP (SHapley Additive exPlanations) to identify and rank the most influential features driving the model's overall decision-making process.
7. To implement local explainability using LIME (Local Interpretable Model-Agnostic Explanations) to provide patient-specific justifications for individual procedure recommendations, thereby fostering clinical trust.

1.6 Research Questions

To address the identified gaps in personalized medical recommendations and model interpretability, this thesis seeks to answer the following research questions:

1. How can structured healthcare data be effectively transformed into a graph-based representation to capture relational similarities between patients for clinical recommendation tasks?

2. To what extent does the integration of a Multi-Layer Perceptron (MLP) for feature extraction enhance the predictive performance of a Graph Attention Network (GAT) in a medical context?
3. What are the optimal hyperparameter configurations for a hybrid graph model regarding neighbor connectivity, attention mechanisms, and learning rates to maximize the accuracy of procedural suggestions?
4. How does the Tuned Hybrid MLP-GAT model compare to a standard baseline GAT architecture in terms of accuracy and F1-score when classifying frequent medical procedures?
5. Which specific patient features (demographic, clinical, or financial) serve as the primary global drivers for medical procedure recommendations within a graph-based framework?
6. How can local explainability methods like LIME be utilized to provide verifiable and clinically relevant justifications for individual AI-generated medical recommendations?
7. To what degree does the application of dual-layered XAI (SHAP and LIME) mitigate the 'black-box' nature of deep learning models in healthcare decision-support systems?

1.7 Scope and Limitations

In this section, we define the boundaries of our research by outlining the specific domains covered within this study and identifying the inherent constraints related to the data, methodology, and technical implementation. Establishing the scope and limitations is

essential for contextualizing the performance of our personalized medical recommendation system and ensuring that our findings are interpreted within the appropriate academic framework.

1.7.1 Scope of the Study

The scope of our research is centered on the development of an interpretable deep learning pipeline for predicting clinical procedures. We specifically focus on the following areas:

- **Dataset Domain:** Our study utilizes synthetic healthcare records generated via the Synthea framework, encompassing 1,183 patients with demographic, procedural, encounter, and medication histories. This allows us to simulate a multi-modal healthcare environment while maintaining data privacy.
- **Algorithmic Framework:** We focus on a Hybrid MLP-GAT architecture. The scope includes the construction of a similarity-based patient graph using K-Nearest Neighbors (KNN) and the comparative analysis of this hybrid model against a baseline Graph Attention Network (GAT).
- **Interpretability Layer:** A primary focus of our scope is the integration of dual-layered Explainable AI (XAI). We implement SHAP for global feature importance and LIME for local patient-level justifications.
- **Procedural Focus:** To ensure statistical reliability and model stability, the scope of our recommendation engine is limited to the top-5 most frequent

procedure categories present in the dataset, such as 'Medication Reconciliation' and 'Assessment of health and social care needs.'

1.7.2 Limitations of the Research

While our research demonstrates robust performance and interpretability, several limitations must be acknowledged:

1. Dataset Limitations: Our research is fundamentally limited by the nature of synthetic data. Although the Synthea dataset provides a structured representation of clinical pathways, it may not fully capture the idiosyncratic complexities, missingness patterns, and noise levels characteristic of real-world Electronic Health Records (EHRs). Furthermore, the sample size of 947 patients after preprocessing, while sufficient for this undergraduate thesis, may not represent the scale required for large-scale clinical deployment [9].

2. Procedural and Class Limitations: Due to significant class imbalance observed in the raw dataset, some of the 36 procedures occurred very infrequently; we restricted our classification task to the top 5 categories. Consequently, our system is currently not designed to recommend rare or specialized medical procedures. The exclusion of minority classes was a necessary trade-off to achieve the final accuracy of 81.58% and avoid model bias toward infrequent labels.

3. Model and Computational Limitations: The Hybrid MLP-GAT model relies on the static construction of a KNN graph. This means the graph structure is determined based on initial feature similarity and does not evolve dynamically during the training process. Additionally, our experiments were conducted within the Google Colab environment using standard GPU acceleration. While this was sufficient for

our architecture, significantly larger graphs with millions of nodes would require more advanced distributed computing resources or graph-sampling techniques like GraphSAGE, which were outside the current implementation scope.

4. Feature and Deployment Limitations: our feature engineering, though comprehensive (encompassing 16 features like 'income_log' and 'age_income_interaction'), does not incorporate unstructured data such as clinical notes, medical imaging, or laboratory time-series data. Furthermore, our study is limited to an offline experimental setting. The real-time deployment of this system into a live Clinical Decision Support System (CDSS) would require additional considerations regarding data latency, API integration, and clinical validation which were not addressed in this research [10].

5. Interpretability Constraints: While SHAP and LIME provide valuable insights, they are utilized as post-hoc surrogate methods. This means they explain the model's behavior rather than making the underlying neural architecture inherently self-explanatory. The interpretations are approximations of the model's logic and should be treated as supportive evidence rather than absolute clinical truths.

By acknowledging these limitations, we provide a transparent foundation for our results and highlight clear pathways for future research aimed at expanding the system's capabilities.

1.8 Significance of the Research

The significance of our research lies in its dual contribution to the fields of computational intelligence and personalized healthcare. By developing a Hybrid MLP-GAT model and integrating it with a robust Explainable AI (XAI) framework, we address a critical bottleneck in the deployment of deep learning models within medical environments. The importance of this study is categorized into its academic, healthcare, and practical significance, each of which is directly supported by the results and methodologies implemented in our research.

1.8.1 Academic and Technical Significance

From an academic perspective, this research contributes to the evolving discourse on Graph Neural Networks (GNNs) in healthcare analytics. While standard architectures often focus on either individual feature processing or relational graph learning, our research demonstrates the technical superiority of a hybrid approach. The development of the Hybrid MLP-GAT model, which combines a Multi-Layer Perceptron for deep feature extraction with a Graph Attention Network for capturing patient-to-patient similarities, represents a significant architectural exploration. Our experimental findings, which show a performance leap from the baseline GAT accuracy of 77.37% to a tuned hybrid accuracy of 81.58%, provide empirical evidence that integrating structured feature learning with attention-based graph connectivity results in more robust predictive models for medical recommendation tasks [11].

1.8.2 Clinical and Healthcare Significance

In the context of healthcare delivery, the significance of our research is found in its potential to enhance the precision of personalized medicine. By constructing a similarity graph of 947 patients through K-Nearest Neighbors (KNN), our system successfully models the relational context of healthcare—recognizing that patients with similar demographic and clinical profiles are likely to benefit from similar medical interventions. This capability is vital for optimizing clinical workflows and ensuring that recommendations, such as 'Medication Reconciliation' or 'Hospice care,' are tailored to the specific historical and economic context of the individual. Our system's ability to process 16 distinct features, including interactions between age and income, ensures that the recommendations are grounded in a multifaceted understanding of the patient's situation [12].

1.8.3 Significance of Explainable AI (XAI)

Perhaps the most impactful aspect of our research is the emphasis on model transparency through XAI. The integration of SHAP and LIME addresses the deep-seated trust issues that often prevent the adoption of AI in clinical settings. The global insights provided by SHAP allow us to validate that our model is prioritizing clinically relevant factors, such as 'unique_procedure' counts and 'AGE,' over irrelevant noise. Simultaneously, the local explanations provided by LIME empower clinicians to see the specific 'why' behind a recommendation for an individual patient. This transparency is not merely a technical addition but an ethical necessity that

ensures AI serves as a reliable assistant rather than an opaque decision-maker. By demonstrating that high accuracy (81.58%) can coexist with high interpretability, we provide a blueprint for safer, more accountable medical AI systems.

1.8.4 Practical Significance and Future Impact

The practical significance of our work extends to the scalability and modularity of the implemented pipeline. The system we have built—covering everything from synthetic data ingestion and feature engineering to graph construction and post-hoc explanation—can be adapted for various clinical departments. Furthermore, by focusing on the top-5 most frequent procedure categories, we have created a stable framework that can serve as a foundation for more complex systems involving rare diseases or specialized surgeries in the future. We believe that our research provides a tangible contribution toward the development of real-time Clinical Decision Support Systems (CDSS). The methodologies explored here offer a clear pathway for reducing diagnostic errors, improving patient outcomes, and fostering a collaborative environment where data-driven insights and human clinical expertise work in tandem.

1.9 Thesis Outline

This thesis is organized into five comprehensive chapters, each detailing a specific phase of our research into personalized medical recommendation systems and explainable artificial intelligence. The structure is designed to guide the reader from the foundational concepts and problem identification through the technical implementation and ultimate evaluation of our proposed framework.

Chapter 1 serves as an introduction to our research. In this chapter, we establish the context of AI in healthcare and define the specific problem of opacity and lack of personalization in current medical recommendation systems. We outline our objectives, research questions, and the significance of developing a hybrid graph-based model. This chapter provides the theoretical and motivational framework that justifies our study and sets the stage for subsequent technical discussions.

Chapter 2 provides a comprehensive review of the existing literature and the theoretical background necessary to understand our work. We examine the evolution of healthcare decision-support systems, the principles of Graph Neural Networks (GNNs), and the emerging field of Explainable AI (XAI). This chapter situates our research within the current academic landscape, identifying the gaps in relational patient modeling and clinical interpretability that our Hybrid MLP-GAT model seeks to address.

Chapter 3 details the research methodology and the technical implementation of our system. Here, we describe the dataset preparation using synthetic healthcare records, the feature engineering process involving 16 distinct attributes, and the construction of the patient similarity graph via the K-Nearest Neighbors (KNN) algorithm. We provide an in-depth explanation of the Hybrid MLP-GAT architecture and the dual-layered XAI integration using SHAP and LIME, outlining the training procedures and hyperparameter optimization strategies employed to achieve our results.

Chapter 4 presents the experimental results and a comparative analysis of our models. We evaluate the performance of the Baseline GAT, the Hybrid MLP-GAT, and the Tuned Hybrid MLP-GAT architectures using metrics such as accuracy, F1-score, and confusion matrices. This chapter also includes the interpretability findings from our XAI analysis, showcasing both global feature rankings and local patient-specific explanations. We discuss the observations derived from our experiments, specifically the 4.21% improvement in accuracy achieved by our final tuned model.

Chapter 5 concludes the thesis by summarizing our primary findings and contributions. We reflect on how our research answered the initial research questions and achieved the established objectives. Furthermore, we discuss the practical implications of our work for clinical decision-making and identify several avenues for future work, including the integration of multi-modal healthcare data and the application of more advanced graph learning architectures like GraphSAGE or Graph Transformers.

1.10 Headings

There are two headings named as B-Head and C-Head are created for preparing the sub-section and sub-subsection of a section. In case of B-head subsection, use ‘Body’ heading to type sub-section information and C-Body for C-Head sub-subsection

Sub-subsection titles will automatically be added to the Table of Contents, directly below the previous sub-section title and with a half-inch indent, each time you update the whole Table of Contents (TOC).

More guidelines are provided in the upcoming chapters of this template. Read them carefully. These guidelines are intended only to help you to use this thesis report template and should not be included in your thesis report.

Chapter 2. Literature Review

The literature review chapter provides the intellectual foundation for your thesis by critically examining existing research and situating your study within the broader academic context. It should not merely summarize prior work but instead synthesize, compare, and evaluate contributions to highlight gaps and justify your research direction [2], [3], [4].

The opening section should explain the purpose of the literature review and how it supports your thesis. Briefly outline the scope of the review, the criteria used for selecting sources, and the organization of the chapter. This helps readers understand the approach you have taken in analyzing prior work.

Provide a broad overview of the field of Computer Science and Engineering relevant to your study. Discuss historical developments, major milestones, and current trends that shaped the research area. This section establishes the context for the more detailed review that follows. Organize the literature either thematically (by topic, method, or perspective) or chronologically (by time).

Move beyond description by evaluating the strengths, weaknesses, and contributions of existing studies. Compare methodologies, datasets, and results, and highlight areas of consensus as well as ongoing debates. This demonstrates your ability to critically engage with the literature rather than simply report it.

Explicitly point out what is missing in the literature. Show how these gaps connect to your research problem and justify the need for your study. This subsection should lead naturally into your research objectives and questions.

If applicable, present the theories, models, or frameworks that underpin your research. Explain how these frameworks guide your methodology and analysis. For example, in machine learning studies, you might discuss the theoretical basis for supervised versus unsupervised approaches.

Define the boundaries of your literature review. Specify what types of studies, timeframes, or methodologies are included, and acknowledge what is excluded. For instance, you may focus only on peer-reviewed journal articles from the last decade or exclude purely theoretical models. This transparency strengthens the credibility of your review.

Explain how the review contributes to your thesis. Highlight its role in advancing understanding of the topic, identifying gaps for future research, and strengthening the rationale for your study. This section should make clear why the literature review is essential to your work.

Table 2.1 An example to understand how to prepare the literature review chapter

Student's Question	Sample Answer
What is it?	The literature review chapter provides the intellectual foundation for your thesis by critically examining existing research and situating your study within the broader academic context. It should not merely summarize prior work but instead synthesize, compare, and evaluate contributions to highlight gaps and justify your research direction [2], [3], [4].
What is its purpose?	It explains the background needed to understand your study, demonstrates familiarity with significant research in your area, and establishes your work as part of the ongoing development of knowledge. It also identifies gaps in the literature and justifies the need for your research.

Student's Question	Sample Answer
What do I need to include?	A broad overview of the field of Computer Science and Engineering relevant to your study, including historical developments, major milestones, and current trends. Include well-established research, the most recent relevant studies, and critical evaluation of methodologies, datasets, and results.
How do I organize it?	Begin with an opening section that explains the purpose of the review, its scope, and criteria for source selection. Then organize the literature either thematically (by topic, method, or perspective) or chronologically (by time). Move beyond description by evaluating strengths, weaknesses, and contributions, highlighting consensus and debates. Explicitly point out gaps that connect to your research problem. If applicable, present theories, models, or frameworks that underpin your study.
What referencing system should I use?	You must follow IEEE Referencing Style for in-text citations and the reference list at the end of your report. Detailed instructions are provided in Section 2.6 of this template.
How do I refer to other authors?	Use IEEE in-text citations (e.g., [2], [3]) within your thesis report. Full reference details must be included in the reference list at the end of your report. See Section 2.6 of this template for details.
How long should it be?	As a rough guide, the literature review should be around 20% of the whole thesis. Your thesis is expected to be about 60% your own work; if the literature review exceeds 40%, it is likely too long. Aim to keep each chapter within 8–10 pages for balance.
What am I supposed to do in my literature review?	You are expected to show that you can recognize relevant and important research in your field, critically evaluate and organize it, and identify gaps that your study will attempt to fill. Demonstrate how existing theories, models, or frameworks guide your methodology and analysis, and explain how your review contributes to advancing understanding of the topic and strengthening the rationale for your study.

In Table 2.1, you will find some of the common questions students ask while writing a literature review, along with suggested answers provided for each.

Some instructions for using this template are discussed below:

2.1 Section Title

Front-part sections, body-part (chapter number and title) sections, and end-part (References) should be formatted using the Section Title style. Section titles will automatically be included in the Table of Contents when you apply the “Section Title” style from the Home tab in the Styles group. However, for them to appear after being first added, you must right-click on the Table of Contents (TOC) and select “Update entire table”.

2.2 B-Head Style

The B-Head style (shown in the Quick Styles menu as “B-Head”) is used as a header for a subsection within a thesis chapter. For example, this subsection title is “B-Head Style,” and if you click on the title, you will see “B-Head” selected in the Styles group. Note: When starting a new chapter, there should always be introductory text before the first use of a B-Head. After a B-Head, apply the Body Text style (shown in the Quick Styles menu as “Body”), which should also be used for the introductory text of the chapter.

B-Heads will automatically be added to the Table of Contents when you apply the “B-Head” style from the Home tab in the Styles group, appearing directly below the chapter title each time you update the entire TOC.

The Body Text style (shown in the Quick Styles menu as “Body”) is used for standard paragraphs. This paragraph is written in Body Text style, and the default style for the next paragraph is also Body Text.

2.3 C-Head Style and Accompanying Paragraph

The C-Head style (shown in the Quick Styles menu as “C-Head”) is used as a header for a sub-subsection within a subsection of a thesis chapter. C-Heads will automatically be added to the Table of Contents when you apply the “C-Head” style from the Home tab in the Styles group, appearing directly below the subsection title each time you update the entire TOC.

2.3.1 Body Text with C-Head

This is a C-Head and its accompanying paragraph. The Body Text style (shown in the Quick Styles menu as “C-Body”) is used only for C-Head section paragraphs. For example, this paragraph is written in C-Body Text style, and the default style for the next paragraph is also C-Body Text.

Note: There should always be introductory text for the subsection before the first use of a C-Head. The style immediately following the C-Head will default to the C-Body Text style, which should be used for the continuation of the sub-subsection.

2.4 Numbered List Style

To create a numbered list, select the lines of text you want to format (whether they are already in a Word list structure or not) and choose the “Numbered List” (shown in the Quick Styles menu as “Numbers”). If the list was previously in a Word list structure, the style will simply update. If the list was formatted manually, check that each item begins and ends correctly and remove any typed numbering.

When a Numbered List style is used, to add a new list item, press Enter at the end of the previous item. To end the list, press Enter and then apply the “Body” style from the Quick Styles menu.

If you wish to create a second numbered list in a different section, follow the directions above. If the numbering does not start from ‘I’, right-click on the first number in the list, select Set Numbering Value, and set it to begin at ‘I’.

2.5 Bulleted List Style

To create a bulleted list, select the lines of text you want to format (whether they are already in a Word list structure or not) and choose the “Bullets” style from the Quick Styles menu. If the list was previously in a Word list structure, the style will simply update. If the list was formatted manually, check that each item begins and ends correctly and remove any typed bullets.

- This is the Bulleted list style.
- To add a list item, press Enter at the end of the previous item.
- To end the list, press Enter and then apply the “Body” style from the Quick Styles menu.

2.6 Referencing

Referencing is the method used to ensure that other research influences are recognized within your practicum report. Correct referencing helps ensure that you avoid being accused of plagiarism. Referencing also ensures that you can demonstrate which sources you have used and how your ideas build upon the research of others. IEEE reference style is required to be used for preparing your practicum report. Your report will be assessed by the supervisor

and practicum committee for ensuring the quality of practicum report referencing. IEEE Referencing includes two main parts:

- Citation within the text of your practicum report (Reference in Text)
- List of references at the end of your practicum report (Reference List)

2.6.1 Referencing in Text

Use sequential numbers in square brackets at the end of the line, but before your punctuation. Use that same number if you are citing the same resource later on in your paper. Bayesian Network is a technique that can be used in safety analysis [1]. Science writing generally doesn't include direct quotations from your sources.

References need not be cited in the text. When they are, they appear on the line, in square brackets, inside the punctuation. Grammatically, they may be treated as if they were nouns or footnote numbers, e.g.: According to [3]; as demonstrated in [2]; as shown by Brown [4], [5]; as mentioned earlier [8], [10], [11], [12], [13], [14], [15]; Smith [4] and Brown and Jones [5]; Wood et al. [7].

Table 2.2 In Text Citation Style.

Citation Style	Example in Text	Notes
Single reference	According to [1].	Reference number appears in square brackets, and inside punctuation.
Sequential references	As mentioned earlier [2], [4], [5], [6], [7], [9].	Multiple references are listed together, and separated by commas.
Non-sequential references	As demonstrated in [2]; as shown by Brown [4], [5].	Different references can be cited separately in the same sentence.

Citation Style	Example in Text	Notes
Author with reference	Smith [4] and Brown and Jones [5].	Author names can be combined with reference numbers.
Group of authors	Wood et al. [7].	“et al.” is used for multiple authors (three or more authors), followed by the reference number.
Treated as noun/footnote	As shown by Brown [4], [5].	Reference numbers act like footnotes or nouns in grammar.

Table 2.2 shows the IEEE Referencing in text quick guidelines. These guidelines will help you to prepare your own thesis report.

2.6.2 The Reference List

A reference list includes details of the sources cited in your paper. It starts on a separate page at the end of your practicum report and is titled References. Each source included in the reference list (after adding the references, select all of the references and click on the RefEntry Style in the quick styles menu) must have been cited in your paper. The following points list the style:

- Reference numbers are set flush left and form a column of their own, hanging out beyond the body of the reference. The reference numbers are on the line, enclosed in square brackets.
- In all references, the given name of the author or editor is abbreviated to the initial only and precedes the last name. Do not use commas around Jr., Sr., and III in names: Michael Smith Jr.; Ray Barnett Sr.; Lucas Molignaro III.

- IEEE publications must list names of all authors, up to six names. If there are more than six names listed, use the primary author's name followed by "et al." For non-IEEE publications, "et al." may be used if additional names are not provided.
- All references, except those ending with URLs, will end with a period, including those with a DOI. However, if a reference contains both a DOI or accessed date, as well as a URL, the DOI or accessed date will be placed first, followed by a period, and then the URL.
- Note that when citing IEEE Transactions, if the issue number or month is not available, research IEEE Xplore to see if the publication carries these details. Note that not all IEEE publications include issue numbers and months.
- Note that early access articles should use the "Date of Publication."
- Articles published within a volume should use the volume and issue publication date.
- All references must include a date that comprises at least the year of publication (or year of the accessed date if applicable). If no date is provided, use the abbreviation "(n.d.)" in the same position where the date would normally be displayed in the reference, preceded and followed by periods, e.g.:
- BAR50 Series Infineon PIN Diode Datasheet. (n.d.). [Online]. Available: <http://www.infineon.com>
- For any reference that cites two months for the same issue, the two months should be separated by a slash (e.g., Jul./Aug.), followed by the year of publication.
- References may not include all information; please obtain and include relevant information.
- Do not combine references. There must be only one reference with each number.

- If there is a URL included with the print reference, it can be included at the end of the reference (see style for online references).

Table 2.3 IEEE Referencing Style quick guide

Type of Reference	How You Should Format It
Journal Article	Format as follows: [sequence-number] Author's first initial. Author's second initial (if provided). Author's last name, (If the reference has additional authors, lists them in the same way; place the word "and" before the name of the last author. If there are seven or more authors, you should list the first author followed by "et al."), "Title of article (within quotation marks, only initial cap first word and proper nouns)," <i>Journal Title</i> (in italics, initial cap the title of the journal in an upstyle), vol. volume-number, no. edition/issue-number, pp. range-of-page-numbers, month abbrev. Month. Year publication-month-year. doi: DOI_number (A Digital Object Identifier (DOI) or URL is highly recommended for online articles if available. If not, you may include a URL preceded by [Online]. Available:). [#] Author's first initial. Author's second initial (if provided). Author's last name, (If additional authors, list them in the same way; place "and" before the last author. For two authors → no comma before "and"; for three or more → use comma before "and". If ≥ 7 authors → list the first author only, followed by et al.), "Title of article (within quotation marks, only initial cap first word and proper nouns)," <i>Journal Title</i> (in italics, initial cap the title of the journal in an upstyle), vol. volume-number, no. issue-number, pp. page range, month abbrev. Year. doi: DOI_number. [#] Author's first initial. Author's second initial (if provided). Author's last name, (If additional authors, list them in the same way; place "and"

Type of Reference	How You Should Format It
	before the last author. For two authors → no comma before “and”; for three or more → use comma before “and”. If ≥ 7 authors → list the first author only, followed by et al.), “Title of article (within quotation marks, only initial cap first word and proper nouns),” <i>Journal Title</i> (in italics, initial cap the title of the journal in an upstyle), vol. volume-number, no. issue-number, pp. page range, month abbrev. Year. [Online]. Available: URL <u>Format: One Author, with DOI and without DOI</u> [#] First Author’s initial(s). Last name, “Title of article,” <i>Journal Title</i>, vol. volume number, no. issue number, pp. page range, Month Year. doi: DOI_number [#] First Author’s initial(s). Last name, “Title of article,” <i>Journal Title</i>, vol. volume number, no. issue number, pp. page range, Month Year. [Online]. Available: URL <u>Example with DOI and without DOI:</u> [1] I. Thompson, “Women and feminism in technical communication,” <i>Journal of Business and Technical Communication</i>, vol. 13, no. 2, pp. 154–178, Apr. 1999. doi:10.1177/1050651999013002003 [2] I. Thompson, “Women and feminism in technical communication,” <i>Journal of Business and Technical Communication</i>, vol. 13, no. 2, pp. 154–178, Apr. 1999. [Online]. Available: https://journals.sagepub.com/home/jbt <u>Format: Two Authors, with DOI and without DOI</u> [#] First Author’s initial(s). Last name and Second Author’s initial(s). Last name, “Title of article,” <i>Journal Title</i>, vol. volume number, no. issue number, pp. page range, Month Year. doi: DOI_number [#] First Author’s initial(s). Last name and Second Author’s initial(s).

Type of Reference	How You Should Format It
	Last name, “Title of article,” <i>Journal Title</i>, vol. volume number, no. issue number, pp. page range, Month Year. [Online]. Available: URL <u>Example with DOI and without DOI:</u> [3] M. S. MacNealy and J. H. Watt, “Strategies for empirical research in writing,” <i>Journal of Technical Writing and Communication</i>, vol. 25, no. 4, pp. 345–360, Oct. 2005. doi:10.1207/s15327625jtwc2504_2 [4] M. S. MacNealy and J. H. Watt, “Strategies for empirical research in writing,” <i>Journal of Technical Writing and Communication</i>, vol. 25, no. 4, pp. 345–360, Oct. 2005. [Online]. Available: https://journals.sagepub.com/home/jtw <u>Format: Three Authors to Six authors, with DOI and without DOI</u> [#] First Author’s initial(s). Last name, Second Author’s initial(s). Last name, and Third Author’s initial(s). Last name, “Title of article,” <i>Journal Title</i>, vol. volume number, no. issue number, pp. page range, Month Year. doi: DOI_number [#] First Author’s initial(s). Last name, Second Author’s initial(s). Last name, and Third Author’s initial(s). Last name, “Title of article,” <i>Journal Title</i>, vol. volume number, no. issue number, pp. page range, Month Year. [Online]. Available: URL <u>Example with DOI and without DOI:</u> [5] R. Smith, T. Jones, and L. Brown, “Deep learning applications in cybersecurity,” <i>Computers & Security</i>, vol. 97, no. 3, pp. 45–60, May 2021. doi:10.1016/j.cose.2021.123456 [6] R. Smith, T. Jones, and L. Brown, “Deep learning applications in cybersecurity,” <i>Computers & Security</i>, vol. 97, no. 3, pp. 45–60, May 2021. [Online]. Available:

Type of Reference	How You Should Format It
	https://www.sciencedirect.com/science/article/pii/S0167404821001234 <u>Format: Seven or more Authors</u> [#] First Author’s initial(s). Last name et al., “Title of article,” <i>Journal Title</i>, vol. volume number, no. issue number, pp. page range, Month Year. doi: DOI_number [#] First Author’s initial(s). Last name et al., “Title of article,” <i>Journal Title</i>, vol. volume number, no. issue number, pp. page range, Month Year. [Online]. Available: URL [7] J. Smith et al., “Advances in machine learning for cybersecurity,” <i>IEEE Transactions on Neural Networks and Learning Systems</i>, vol. 32, no. 5, pp. 1234–1248, May 2021. doi:10.1109/TNNLS.2021.123456 [8] J. Smith et al., “Advances in machine learning for cybersecurity,” <i>IEEE Transactions on Neural Networks and Learning Systems</i>, vol. 32, no. 5, pp. 1234–1248, May 2021. [Online]. Available: https://ieeexplore.ieee.org/document/123456
Conference Paper	<u>A more accurate and complete IEEE format for an online published conference paper with a DOI (the most common modern format) is:</u> [#] Author’s first initial. Author’s second initial (if provided). Author’s last name, (If additional authors, list them in the same way; place “and” before the last author. For two authors → no comma before “and”; for three or more → use comma before “and”. If ≥ 7 authors → list the first author only, followed by et al.), “Title of paper in sentence case,” <i>Abbreviated Title of Conference</i>, Month Year, pp. page range, doi: DOI_number <u>If the proceedings were published in print and no DOI is available, the format includes location and publisher details:</u>

Type of Reference	How You Should Format It
	[#] First Author's initial(s). Second Author's initial(s). Last name, (Additional authors listed the same way; use "and" before the last author. For two authors → no comma before "and"; for three or more → comma before "and". If ≥ 7 authors → first author only, followed by et al.), "Title of paper in sentence case," in <i>Abbreviated Title of Conference</i>, City, State, Country: Publisher, Year, pp. page range. <u>IEEE Format for Online Published Conference Paper (with DOI):</u> [#] Author(s) initial(s). Surname(s), "Title of paper in sentence case," <i>Abbreviated Title of Conference</i>, Month Year, pp. page range, doi: DOI_number. [1] G. R. Faulhaber, "Design of service systems with priority reservation," in <i>Conf. Rec. 1995 IEEE Int. Conf. Commun.</i>, Jun. 1995, pp. 3–8, doi:10.1109/ICC.1995.123456. [2] A. Kumar and B. Rahman, "Blockchain-based IoT security framework," in <i>Proc. IEEE Int. Conf. on Communications (ICC)</i>, Jun. 2022, pp. 123–130, doi:10.1109/ICC.2022.9812345. [3] R. Smith, T. Jones, and L. Brown, "Deep learning applications in cybersecurity," in <i>Proc. IEEE Int. Conf. on Computer Communications (INFOCOM)</i>, May 2021, pp. 456–462, doi:10.1109/INFOCOM.2021.9487654. [4] J. Smith et al., "Advances in machine learning for cybersecurity," in <i>Proc. IEEE Int. Conf. on Neural Networks (IJCNN)</i>, Jul. 2021, pp. 789–796, doi:10.1109/IJCNN.2021.9532147. <u>IEEE Format for Print-Published Conference Paper (no DOI):</u> [#] Author(s) initial(s). Surname(s), "Title of paper in sentence case," in <i>Abbreviated Title of Conference</i>, City, State, Country: Publisher, Year, pp. page range.

Type of Reference	How You Should Format It
	[1] I. Thompson, "Women and feminism in technical communication," in <i>Proc. IEEE Int. Conf. on Professional Communication</i>, Boston, MA, USA: IEEE, 1999, pp. 154–178. [2] A. Kumar and B. Rahman, "Blockchain-based IoT security framework," in <i>Proc. IEEE Int. Conf. on Communications (ICC)</i>, Paris, France: IEEE, 2022, pp. 123–130. [3] R. Smith, T. Jones, and L. Brown, "Deep learning applications in cybersecurity," in <i>Proc. IEEE Int. Conf. on Computer Communications (INFOCOM)</i>, New York, NY, USA: IEEE, 2021, pp. 456–462. [4] J. Smith et al., "Advances in machine learning for cybersecurity," in <i>Proc. IEEE Int. Joint Conf. on Neural Networks (IJCNN)</i>, Tokyo, Japan: IEEE, 2021, pp. 789–796.
Book	<u>IEEE Book Reference Format: Online Book (with DOI or URL):</u> <u>Template:</u> [#] A. Author, B. Author and C. Author (or A. Author et al. if ≥ 7 authors), <i>Book Title</i>, xth ed. (if applicable). City, State (if USA) or Country: Publisher, Year. doi: DOI_number <u>Example:</u> [1] M. S. MacNealy and J. H. Watt, <i>Strategies for Empirical Research in Writing</i>, 2nd ed. Boston, MA, USA: Allyn and Bacon, 1999. doi:10.4324/9781315853178 <u>If no DOI but available online:</u> <u>Template:</u> [#] Author's first initial. Author's second initial (if provided). Author's last name, <i>Book Title</i>, xth ed. (if applicable). City, State (if USA) or Country: Publisher, Year. [Online]. Available: URL

Type of Reference	How You Should Format It
	Example: [1] J. H. Watt and S. A. van den Berg, <i>Research Methods for Communication Science</i>, Boston, MA, USA: Allyn and Bacon, 1995. [Online]. Available: https://archive.org/details/researchmethodscomm <u>Print Book (no DOI)</u> <u>Template:</u> [#] Author's first initial. Author's second initial (if provided). Author's last name, <i>Book Title</i>, xth ed. (if applicable). City, State (if USA) or Country: Publisher, Year. <u>Example:</u> [1] M. S. MacNealy, <i>Strategies for Empirical Research in Writing</i>, Boston, MA, USA: Allyn and Bacon, 1999.
Website	<u>IEEE Website Reference Format: Template:</u> [#] Author(s) Initial(s). Surname(s) (or Name of Organization), "Title of page in sentence case," Website Name, [Online]. Available: URL (Accessed Abbrev. Month Day, Year). <u>Example: Organization as Author</u> [1] IEEE, "Author guidelines for conference papers," IEEE, [Online]. Available: https://www.ieee.org/conferences/publishing/templates.html (accessed Dec. 4, 2025). <u>Example: Individual Author</u> [2] J. Doe, "Introduction to quantum computing," Quantum Research Hub, [Online]. Available: https://www.quantumresearchhub.org/intro (accessed Dec. 4, 2025). <u>Example: News Website</u>

Type of Reference	How You Should Format It
	[3] BBC News, "Global cybersecurity threats rise in 2025," BBC News, [Online]. Available: https://www.bbc.com/news/cybersecurity-2025 (accessed Dec. 4, 2025).
If your reference source is different or your reference scenario is not covered in this table: Check the official IEEE reference guide .	

Table 3.3 shows the IEEE Referencing Style quick guidelines. These guidelines will help you to prepare your own practicum report references.

Chapter 3. Research Methodology

The methodology section follows immediately after the literature review in your thesis report and should flow naturally from it. By this stage, you will have defined your research question and conducted a detailed review of existing scholarship on your topic. You will also have examined how other scholars arrived at their conclusions, the assumptions underlying their work, the theoretical frameworks they employed, and the methods they used to collect, organize, and present their data. Drawing on these observations, as well as discussions with your supervisor, you will have planned how to address your own research question. This may involve determining how you will gather data, identifying the models you will use to process it, or clarifying the philosophical positions that inform your work. Your methodology section should therefore provide a detailed account of both how you intend to approach your dissertation and why you have chosen this particular approach.

The methodology section may include the following components: Recap of research questions, Description of design or method, Background and rationale for design choice, Evaluation of method and statement of limitations, Data sources and materials, Sampling and participants (if applicable), Ethical considerations, Procedures and workflow, Data analysis techniques, and Validity and reliability (or trustworthiness in qualitative research). Details of the these components are discussed below:

I. Recap of Research Questions

When introducing your methodology, begin by revisiting the central research problem or questions you posed at the outset. The methodology should be designed to address these directly. Rather than repeating the questions verbatim, reframe them in a way

that connects your literature review to the methodological choices you are about to present. This creates a logical bridge between what has been established in prior studies and how your approach will generate answers.

II. Description of Design or Method

Provide a clear and detailed account of the process you followed to gather and analyze data, or to address your research question. Ensure transparency so that another scholar could understand, evaluate, and replicate your approach in a different context.

III. Background and Rationale for Design Choice

Justify why you selected this method. Ground your rationale in the literature review, showing that your choice is informed by established scholarship while also reflecting originality. Explicitly connect the rationale to your research problem, demonstrating that the methodology is a tailored response to the questions you aim to answer.

IV. Evaluation of Method and Statement of Limitations

Critically assess the strengths and weaknesses of your chosen approach. Acknowledge trade-offs honestly but without apology, and justify why the method remains the most appropriate for your study despite its limitations.

V. Data Sources and Materials

Specify the datasets, documents, tools, or instruments used. This ensures transparency and reproducibility.

VI. Sampling and Participants (if applicable)

Describe how participants or cases were selected, including inclusion/exclusion criteria. This adds credibility to qualitative or quantitative studies.

VII. Ethical Considerations

Outline how you addressed consent, confidentiality, and compliance with institutional or disciplinary ethics standards. This section is often mandatory.

VIII. Procedures and Workflow

Provide a step-by-step account of how the research was conducted (e.g., experiment setup, survey distribution, simulation runs, etc.). This helps readers follow your process clearly.

IX. Data Analysis Techniques

Explain the statistical, computational, or qualitative methods used to interpret the data. Show alignment between your research questions and the chosen analytical tools.

X. Validity and Reliability (or Trustworthiness in qualitative research)

Discuss how you ensured accuracy, consistency, and robustness of your findings. This strengthens the credibility of your methodology.

The rest of this section shows how to use this template to prepare the figures and tables in your thesis report.

3.1 Figures

To insert a figure, place your cursor at the beginning of the Figure Title line. From the Insert ribbon, choose Text Box and draw a text box in the desired location. Place the image inside the text box. Once the image is inserted, click on the picture and open the dropdown menu that appears. Make sure the “In Line With Text” option is selected to keep the figure properly aligned and associated with the correct text.

For Example: The box shown below is a placeholder to demonstrate how the figure should appear.

You can place a report figure inside a text box.

Figure 3.1 Example Figure Title.

Place the figure 3.1 description here by applying the Figure Description style (see the Section 3.1.2, Figure Description Style).

3.1.1 Figure Style

The Figure Title style, which appears in the Quick Styles menu as “FigureCell” is used to name a figure in a way that will appear in the List of Figures. The title, along with its page number, will automatically be added to the List of Figures when you update the entire table.

3.1.2 Figure Description Style

The Figure Description style (shown in the Quick Styles menu as “FigDesc”) is used to briefly describe the accompanying figure. The spacing after the description separates it from the body text that follows below the figure.

You can place a report figure inside a text box.

Figure 3.2 Example Figure Title2.

The Figure Title style and Figure Description style have been applied to present the Figure 3.2 and its accompanying description.

3.2 Tables

Tables can be copied and pasted from another document or created directly in Word using the Insert Table command on the Insert tab of the ribbon.

After inserting a table by either method, apply the following styles:

3.2.1 Table Title Style

The Table Title style (shown in the Quick Styles menu as “TableTitle”) should be applied immediately before the table itself. This ensures the title is properly formatted and included in the Table of Contents.

3.2.2 Table Cell Style

To apply this style to the table cells, highlight the table and choose “Table Cell” from the Quick Styles menu. By default, text inside the cells will be left-aligned. To center text (as in the second and third columns above), select those cells and choose Center. For vertical

alignment, select the applicable cells, go to the Layout tab in the ribbon, and in the Alignment section on the right, choose the graphic that matches your preferred alignment (top left, bottom right, or centered both horizontally and vertically).

3.2.3 Table Description Style

The Table Description style (shown in the Quick Styles menu as “TableDesc”) is used to format the brief description of the table immediately following it.

A sample table with description style has shown below:

Table 3.1 Table Title.

Process	Arrival Time	Service time
A	0	4
B	3	5

Place the description for Table 3.1 here using the Table Description style. (e.g., Table 3.1 shows the arrival time and service time for the process A and process B.).

Chapter 4. Results and Discussion

The Results and Discussion chapter is the heart of your thesis, where raw data is transformed into meaningful insights. It presents your findings, explains their significance, and connects them to your research aims, hypotheses, and existing literature. Unlike the methodology or literature review, this chapter focuses on evidence and interpretation, demonstrating the value of your research in Computer Science and Engineering.

4.1 Presenting Results

Results should be presented in a clear, logical, and structured manner. Use tables, figures, and graphs to illustrate key outcomes, ensuring they are properly labeled and referenced. Report results objectively, without interpretation at this stage. Organize them in a way that best supports your study, such as:

- I. From most important to least important (e.g., from statistically most significant to least significant).
- II. By research questions or problems, presenting results that directly answer each question sequentially.
- III. By methods or experimental elements, grouping procedures with their corresponding results.

Summarize data rather than overwhelming readers with raw outputs, and highlight findings that directly address your research objectives.

4.2 Interpreting Results in Discussion

After presenting the data, move into the discussion where you interpret and explain what the results mean. Relate them directly to your research aims and hypotheses, showing whether they confirm, extend, or challenge expectations. Compare your findings with prior studies and theoretical frameworks, highlighting similarities, differences, and possible reasons for discrepancies. This critical analysis situates your work within the broader academic context.

4.3 Connecting to Research Objectives

Explicitly connect your discussion back to the objectives and research questions stated earlier in the thesis. For each objective, explain how your findings provide answers or insights. This alignment reinforces coherence and demonstrates that your study has achieved its intended goals.

4.4 Addressing Limitations

A strong discussion acknowledges limitations. Reflect on constraints such as dataset size, simulation environment, or methodological assumptions that may have influenced the results. Transparency about limitations demonstrates academic integrity and provides direction for future research.

4.5 Highlighting Contributions and Implications

Discuss the significance of your findings. Explain how they advance theoretical understanding, improve practical systems, or have potential industrial or societal

applications. For example, if your work improves algorithm efficiency, highlight its impact on real-world computing or communication systems.

4.6 Structuring the Chapter

You may choose to:

- Separate results and discussion into distinct sections, or
- Integrate them together, presenting each finding followed immediately by its interpretation.

The choice depends on your discipline and supervisor's guidance, but clarity and logical flow are essential.

The rest of this chapter provides some additional styles that you are going to use in your thesis report. Styles used in other parts of the document:

4.7 Thesis Title Style

The ThesisTitle style is used on the first page and second page of the document to format the thesis title.

4.8 Department School University Style

The Dept-College-Unv. style is applied on the first page and Second page to format the departmental information, ensuring it appears centered and distinct from other text.

Chapter 5. Conclusion

The Conclusion chapter is the final section of a thesis and serves as the place where the researcher ties together all elements of the study. It should not introduce new data or arguments but instead provide a reflective closure that synthesizes the research problem, findings, contributions, limitations, and future directions. Students should approach this chapter as both a summary and a forward-looking statement, ensuring that it is concise, cohesive, and academically rigorous. The goal is to leave the reader with a clear understanding of the research impact and its relevance to the broader field of Computer Science and Engineering.

5.1 Restating the Research Problem and Objectives

Begin by revisiting the central research problem and the objectives that guided the study. Present them as a reflective summary that frames the research in light of what has been accomplished. This establishes continuity and prepares the reader to understand the significance of the findings.

5.2 Summarizing Key Findings

Provide a concise synthesis of the major findings. Highlight how the research questions were answered or how the objectives were met. Focus on the broader implications of the results rather than technical detail or raw data.

5.3 Highlighting Contributions

Clearly articulate the contributions of the thesis. These may be theoretical, methodological, or practical. State explicitly how the work adds value to the academic community and industry, and distinguish it from existing literature.

5.4 Discussing Limitations

Include a paragraph that acknowledges the limitations of the study. Discuss constraints such as scope, resources, or assumptions. Maintain a balanced tone that recognizes limitations without undermining the overall contribution.

5.5 Suggesting Future Work

Offer realistic and specific directions for future research. In Computer Science and Engineering, where technology evolves rapidly, suggestions might include scaling the proposed solution to larger datasets, testing in real-world environments, or integrating with emerging technologies. The future work section should inspire subsequent researchers while remaining grounded in the scope of the thesis.

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Plagiarism Report

(Plagiarism should be less than 25%)

Attach the proof here.